

Mahmoud Alyosify

Specialised MLOps & Cloud AI Engineer | Model Deployment, Pipelines & Cost-Efficient Inference | MSc Artificial Intelligence, Queen's University (Canada)

✉ mahmoud.alyosify@gmail.com | 📞 +20 114 555 7959 | 📍 New Cairo, Egypt · Open to Relocation
🌐 [linkedin.com/in/mahmoudalyosify](https://www.linkedin.com/in/mahmoudalyosify) | 🐙 github.com/MahmoudAlyosify | 📁 Portfolio | 🤖 Hugging Face | 📊 Kaggle

PROFESSIONAL SUMMARY

MLOps & Cloud AI Engineer focused on the gap between a trained model and a system that runs reliably and affordably. Deployed **fine-tuned LLMs and vision encoders** to **AWS** via **ONNX, Docker and FastAPI**; built **deterministic, reproducible benchmarking** on pinned GPU instances measuring **TTFT, inter-token latency and GPU energy** through **NVML**; and trained **deep-RL agents** that **provision cloud resources against an explicit cost-latency budget**. Holds **three AWS Academy credentials** (ML Foundations, ML for NLP, Data Engineering). Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**, with prior production-systems experience on **national-scale toll infrastructure**.

EDUCATION

- MSc in Artificial Intelligence
Queen's University, Kingston, Ontario, Canada

Sep 2025 – Nov 2026
Expected Nov 2026

 - **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
 - Coursework: **Cloud Computing, End-to-End ML Pipelines, Deep Learning, Large Language Models, Reinforcement Learning.**
- B.Sc. Computer & Information Science (Bioinformatics)
Assiut University, Egypt

Sep 2019 – Jul 2023
GPA: 3.53 / 4.00

 - Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023.**

TECHNICAL SKILLS

Cloud & Infrastructure	AWS (GPU Instances, ML & Data Services), Docker, CI/CD, Linux, Bash, Infrastructure Cost Optimisation, Autoscaling Strategies, Self-Hosted Model Serving
Model Serving & Deployment	ONNX, FastAPI, Streamlit, FAISS, ChromaDB, Containerised Inference, Batch & Interactive Serving, API Design
Reliability & Measurement	Deterministic & Reproducible Pipelines, Pinned Environments and Seeds, Latency Profiling (TTFT, Inter-Token), Throughput Benchmarking, GPU Energy Monitoring (NVML), Baseline Comparison
Data Engineering	ETL Pipelines, PySpark, Pandas, NumPy, SQL, Data Validation, MySQL, MongoDB, SQL Server
ML & AI	PyTorch, TensorFlow, Scikit-learn, Hugging Face, Fine-Tuning (LoRA, QLoRA, PEFT), Reinforcement Learning (PPO, DQN), Optuna
Programming	Python, SQL, C#, C++, C, Go, JavaScript, Bash

SELECTED DEPLOYMENT & INFRASTRUCTURE PROJECTS

- RL-Cloud-Autoscaler — Learned Cloud Resource Provisioning
Python · PPO · DQN · Gymnasium · Cloud Cost Optimisation

2026

 - Built a **custom Gymnasium environment** simulating cloud workload dynamics, then trained **PPO** and **DQN** agents to make provisioning decisions in real time.
 - Optimised an explicit **cost-latency reward** and benchmarked against **static allocation** and **reactive threshold-based autoscaling** — the policies teams actually run — to quantify real savings rather than simulated ones.
- Verbosity-Aware Decoding — Reproducible GPU Benchmarking Harness
MSc Research Project (CISC 898), Queen's University · AWS GPU · NVML

2026 – Present

 - Built a fully **deterministic, reproducible benchmarking pipeline** on **pinned AWS GPU instances** with fixed seeds and versioned environments, so efficiency claims are verifiable rather than anecdotal.
 - Instrumented **TTFT, inter-token latency** and **GPU energy via NVML** alongside token counts, across three model families and four benchmarks.
 - Delivered a **zero-overhead inference optimisation** — no retraining, no architecture change, no additional KV-cache or attention cost — deployable as a drop-in component.
- Horus-OSINT — Fine-Tuned LLM Deployed on AWS
Meta-Llama-3-8B · Fine-Tuning · AWS · Self-Hosted Serving

2026

 - Owned the full path from a **159,826-record** dataset through fine-tuning to a running **AWS-hosted** interactive service — data preparation, training, packaging and deployment.
- SimCLR-Vision-SSL — ONNX Export & Vector-Search Serving
PyTorch · ONNX · FAISS · Retrieval

2026

 - Exported a trained encoder to **ONNX** and served embeddings through a **FAISS** index — converting a research model into a framework-independent, deployable retrieval component.

Additional Projects

- **HORUS Sentinel** — multi-agent platform with a self-hosted fine-tuned LLM and knowledge-graph state layer running continuously.
- **Automated Multimodal Agent** — multi-stage automation pipeline chaining extraction, generation and synthesis services.
- **Also:** SCRAP · AudioShield (Streamlit-served CNN) · Human vs. Agentic Pull Requests · Vitalism Solution.

PROFESSIONAL EXPERIENCE

- Systems Engineer (Military Service)**
National Company for Roads Building & Development

Feb 2024 – Mar 2025
Cairo, Egypt

 - Supported production backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**, where downtime had immediate operational cost.
 - Debugged and optimised real-time transportation software and supported legacy applications to maintain continuity; co-developed an internal data-management and reporting application in **C#, .NET, Entity Framework, SQL Server** and **Crystal Reports**.
- Machine Learning Instructor**
National Telecommunication Institute (NTI)

Jul 2025 – Sep 2025
Cairo, Egypt

 - Delivered a **90-hour** hands-on ML program covering supervised learning, PCA and neural networks with practical labs.
- AI in Cybersecurity Intern**
e& Egypt — via NTI (On-the-Job Training)

Jun 2025 – Jul 2025
Cairo, Egypt

 - Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.
- Online Instructor — AI, Algorithms & Data Science**
Udemy & YouTube

2020 – Present
Remote

 - Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

CERTIFICATIONS & AWARDS

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| AWS Academy — Data Engineering | AWS Academy — ML Foundations, ML for NLP |
| Machine Learning Specialization — DeepLearning.AI | HCIP – AI (Huawei) |
| Full Stack Diploma — .NET & Angular (Route Academy) | Microsoft Certified: Data Analyst Associate |
| 1st Place — Smart Cities Hackathon 2022 | Semi-Finalist — Microsoft Imagine Cup 2023 |
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LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.
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